

NAVRAJ SINGH

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EDUCATION & PROFESSIONAL TRAINING

Indian Institute of Science Education and Research (IISER) Mohali

Punjab, India

BS-MS Dual Degree Student (Fifth Year)

Oct 2022 – Present

CPI: 9/10

Note: Detailed reports and code for the projects available at [Github](#).

AI SAFETY & ALIGNMENT RESEARCH

Evaluation Awareness in Large Language Models

Pre-print on arXiv

Lead Researcher | Supervisor: Maheep Chaudhary

Feb – June 2026

- Mapped the representational scaling of evaluation awareness across compact architectures (Gemma 3, Phi-3, Llama-3 8B). Utilized Chain-of-Thought analysis, representation probing, and Integrated Gradients attribution for the investigation.
- Engineered a dual-pathway mitigation pipeline combining automated prompt sanitization with activation counter-steering, to neutralize evaluation artifacts across 200 high-confidence prompts, achieving an average behavioral flip rate of 70.58% across all evaluated architectures.

Interpretability Frameworks for Diffusion Models

Ongoing MS Thesis

Research Student | Supervisor: Prof. Ashutosh Modi

May 2026 – Present

- Building mechanistic evaluation tools to analyze feature representation and concept formulation during the reverse denoising process.
- Applying the Schrödinger bridging framework to track multi-step trajectories from Gaussian noise to structured outputs, identifying precise layers where early generation choices stabilize.

Narrow Secret Loyalty Extension Using CAI

Research Fellow | BlueDot Impact Technical AIS Project | USD1,600 research grant June - August 2026

- Extended Lamerton & Roger (2026) via SL-CAI and RL-CAI pipelines on a 7B Qwen-2.5-Instruct model to probe for hidden loyalty behaviors under adversarial affordances.
- Found action-breadth entropy increased monotonically across baseline $\rightarrow$ SL-CAI $\rightarrow$ RL-CAI, and intent selectivity improved substantially after SL-CAI. Black-box detection rates stayed statistically indistinguishable from baseline across most affordance levels which is evidence that standard black-box audits can miss loyalty shifts that are present internally.
- Submitted to Apart Research's Secret Loyalties Sprint. LW write-up in progress.

Bluedot Technical AI Safety Course

Scholar – 6-Day Intensive Cohort

June 2026

- Selected for a technical training cohort covering alignment theory, mechanistic interpretability foundations, and frontier model vulnerability profiles.

VEIL: Mechanistic Upstream Guardrails for Generative Biology

Apart Research Hackathon

Core Developer | AIBio Track

- Trained an L_1 -regularized linear probe on ProteinMPNN decoder latents to identify structural markers of pathogenic risk, isolating causal features independent of token length.
- Analyzed feature polysemanticity and exposed generalization boundaries on unseen biological targets (e.g., Ricin) where safety vectors conflated structural threat with structural rigidity.

Mechanistic Analysis of Simplicity Bias and Feature Steering

PRECOG Lab

Independent Researcher

- Trained Sparse Autoencoders (SAEs) on penultimate activations to disentangle compressed neural features.
- Proved a mechanistic variance between simple and robust features via targeted feature steering interventions, applying token-level gradients to successfully manipulate downstream classifications.

Replication Studies

- Completed replication studies of various foundation papers in the field of Foundational DL and AI Safety.

OTHER TECHNICAL & QUANTITATIVE INTERNSHIPS

Fully Homomorphic Encryption in AI (CKKS)

Supervisor: Prof. Mridula Verma

Jul 2025 – Nov 2025

- Implemented encrypted neural networks running evaluations directly on ciphertexts to ensure strict data privacy.
- Managed CKKS noise budgets and scaling factors across deep architectures to optimize the precision-compute tradeoff.

Valid Circuit Detection in Hexa-board PCB

Supervisor: Prof. Satyajit Jena

May 2025 – Jul 2025

- Engineered an end-to-end computer vision object-detection pipeline leveraging a hybrid SSD architecture (ResNet-50 backbone paired with an auxiliary ResNet-18 classifier) for multi-scale anomaly detection.

Computational Methods for Option Pricing

Supervisor: Prof. Anindya Goswami

Jun 2024 – Aug 2024

- Developed high-performance C++ implementations of continuous-time option pricing and stochastic algorithms.

Grain Size Distribution Analysis Using Mask-RCNN

Supervisor: Prof. Yunus Ali Pulpadan

Jun 2024 – Aug 2024

- Fine-tuned a custom Mask-RCNN framework on drone telemetry to segment and calculate granular distribution frequencies.

TECHNICAL SKILLS

- **Languages & Libraries:** Python, C++, PyTorch, TensorFlow, CUDA, R, NumPy, Scikit-learn, Pandantic, aiosqlite.
- **Core Competencies:** Mechanistic Interpretability, Sparse Autoencoders (SAEs), Circuit Discovery, Linear Probing & Steering, Causal Interventions (Activation Patching), Fully Homomorphic Encryption (CKKS).
- **Mathematics:** Linear Algebra, Measure Theoretic Probability, Cryptography & Number Theory, Group Theory, Stochastic PDEs.

ACHIEVEMENTS & RELEVANT COURSEWORK

- **Achievements:** JEE Main (Top 1 Percentile), IISER Aptitude Test (AIR-1345), KVPY Written Test (Qualified), SOF Science Olympiad (State Rank 51), SOF Mathematics Olympiad (State Rank 102).
- **Key Courses:** Machine Learning (Stanford CS229), Deep Learning, Quantum Information & Computation, Theory of Computation, Measure Theoretic Probability, Functional Analysis, Non-Linear Dy-

namics.